Comparing False Crawls and Pseudo-Absence Points when Modeling Sea Turtle Nesting on Pensacola Beach

Ashley Ordonez

2025-12-05

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Abstract

Objectives

Loggerhead sea turtles (*Caretta caretta*) select nesting sites based on various environmental factors, yet the influence of beach morphology on nesting decisions remains unclear. In order to model nesting patterns, both observed nests and non-observed nests must be present in the data. The absence of nests can be represented with either pseudo absence points, which are randomly selected via ArcGIS, or false crawl points, which are observed and recorded by state employees while recording observed nests.

This study evaluated how different representations of "non-nesting" locations affect model estimation and classification performance when modeling loggerhead sea turtle (Caretta caretta) nesting on Pensacola Beach. Specifically, we assessed whether binary and multinomial logistic regression models produced different inferences about beach slope and foreshore slope on nesting likelihood, and whether the choice of absence data influenced prediction accuracy.

Methods

Presence (P) and false crawl (FC) locations were recorded by trained observers along Pensacola Beach, while pseudo-absence (PA) points were generated in ArcGIS by randomly sampling areas outside known nest buffers. Beach slope and foreshore slope were then calculated for each location.

Three models were fit:

1. Binary logistic regression, comparing P vs. PA.
2. Binary logistic regression, comparing P vs. FC.
3. Multinomial logistic regression, simultaneously comparing P vs. PA and P vs. FC.

For each model, adjusted odds ratios (ORs), 95% confidence intervals, and Wald z tests assessed the significance of foreshore slope and beach slope. Statistical significance was defined as $p < 0.05$.

To examine classification, predicted probabilities were used to generate confusion matrices, which identify the number of true positive, true negative, false positive, and false negative classifications. Using the entries in the confusion matrix, we calculated accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV).

Results

Slope variables had small, non-significant effects across all modeling approaches. In binary models, a 1° increase in beach slope decreased the odds of nesting by 5% when comparing to FC but incrased the odds of nesting by 2% when comparing to PA. Foreshore slope increased odds of nesting by 1–5%, depending on model, but none of these effects were significant ($p > 0.15$ for beach slope; $p > 0.45$ for foreshore slope). Multinomial models produced similar estimates and identical inferential conclusions.

Classification performance differed by absence type. Pseudo-absence models achieved moderate accuracy (56%) and modest sensitivity (0.55–0.65). Interestingly, False-crawl models failed to classify FC points as such, producing specificity near 1 and sensitivity of 0. Across all cases, binary and multinomial approaches yielded nearly identical accuracy, PPV, NPV, and classification patterns.

Conclusions

The choice between pseudo-absence points and observed false crawls affects model performance, but does not materially change estimated effects of environmental predictors. Beach slope and foreshore slope showed weak, non-significant associations with nesting probability, regardless of modeling approach. Because false crawls were never classified correctly, pseudo-absence points produced more stable and interpretable results for presence-absence modeling in this application. Both binary and multinomial logistic regression generated similar conclusions, suggesting little added benefit to using the more complex multinomial logistic regression model.

1 Introduction

1.1 Statement and Relevance of Problem

There is a small population of loggerhead sea turtles that choose to nest on Pensacola Beach. There are many studies that have looked into the nesting patterns of the loggerhead population else in Florida, but few have examined specifically within the Florida Panhandle. This research is part of an effort to fill in the gaps for the Florida Panhandle area.

The Florida Fish and Wildlife Conservation Commission (FWC) perform “turtle patrol” during nesting season and record observed turtle nests, however, it is difficult to determine where they choose not to nest. In order to observe an “absence,” researchers can use observed false crawls or computer-generated pseudo-absence points. A false crawl is when a turtle comes ashore but does not nest. Pseudo-absence points are randomly generated locations on the beach where no nest is observed. Both methods have been used in previous studies, but there is a lack of research comparing the two as the “no nest” observation.

In this study, we are looking to determine how false crawl and pseudo absence points compare as the “no nest” observation. We are also comparing multiple binary logistic regressions to a single multinomial logistic regression to determine, between the two options for modeling methods, if one stands out as a “better” model.

1.2 Literature Review

The Computational Geomorphology and Modeling Lab at UWF has been investigating nest sites of loggerhead sea turtles on Pensacola Beach for several years. In 2022 Jordan Iserman conducted the study “Beach Morphology Characteristics of Sea Turtle Nesting Sites: A Statistical Analysis” in which they compared the beach characteristics within 100 meters of a nesting site (1). For this study they only evaluated the beach characteristics of observed nests. The results of their Wilcoxon signed rank showed a difference in foreshore slope and beach slope (1).

Aiden Jensen conducted the study “An Exploration of Presence and Pseudo-Absence Data in the Analysis of Loggerhead Sea Turtle Nesting Behavior in the Florida Panhandle” in 2022 (2). The study was a pseudo simulation in which the dataset for the 10:1 ratio was resampled to generate different ratios of pseudo absence points to presence points. They used logistic regression to evaluate bias across ratios and found that as the number of psuedo absence points increased, the bias also increased (2).

In 2024, Cheyenne Long did “Modeling Loggerhead Nesting Patterns: How Many Pseudo-Absence Points are Necessary?” by doing a full Monte Carlo simulation study (3). They looked at 1:1, 2:1, 5:1, and 10:1 ratios and were able to specify the true relationships, then examined bias and mean square error of the resulting models. Like in Jensen’s results, their results showed an increased bias for β_i as the pseudo-absence point ratio increased (3).

Most recently, Angelina Scamardo worked on “From the Sea to Statistics: Using Machine Learning to Predict Sea Turtle Nesting Patterns” in which they compared machine learning methods to examine accuracy in predicting nesting status between presence and pseudo-absence points (4). The results of the study found that the random forest approach had higher accuracy, sensitivity, and specificity than the support vector machine approach (4).

Outside of Pensacola, there are several researchers who have conducted similar studies on the loggerhead population. Afford did the “Using multivariate analysis to determine characteristics of sea turtle nest selection along the Florida Panhandle” study in 2016 (5). They examined the nest selection of loggerhead sea turtles in Okaloosa county using pseudo absence points as the “no nest” sites. Using Bayesian logistic regression, classification trees, and random forest analyses they looked to identify environmental predictors of nesting (5).

Most recently, Hernandez conducted “Sea Turtle Nesting on Nourished Beaches With Different Construction Designs: A Case Study in Southeast Florida, USA” in which they examined nest selection of loggerhead and green sea turtles in South Florida (6). They used false crawl points as their “no nest” sites and employed descriptive statistics and zone-based proportions to compare, rather than inferential statistics (6).

Byrd (2022) examined environmental and anthropogenic drivers of loggerhead sea turtle false crawls on Jekyll Island, Georgia, using historical nesting records (2008–2019) and binary logistic regression analyses on contemporary field data (2020–2021) (7). They evaluated the effects of revetment proximity, lighting, sand temperatures, human presence, and beach obstructions on nesting status using false crawls as their “no nest” sites. Byrd found that false crawl rates were consistently and significantly higher near the large shoreline revetment, with additional context-specific effects from subsurface sand temperature and human presence, indicating that major anthropogenic structures are the primary contributors to elevated false crawling on this beach (7).

Manestar conducted a study examining the influences on the nest site selection of loggerhead sea turtles using unmanned aerial vehicle surveys and traditional methods in Boca Raton, Florida (8). With false crawl points as their “no nest” sites as well, they used logistic regression to identify predictors of nesting (8).

2 Data and Methods

2.1 Data Description

The data used for this study has a categorical outcome for nest type in which the possible outcomes are as follows:

$$\text{Nest Type} = \begin{cases} \text{P} & \text{Presence} \\ \text{PA} & \text{Pseudo Absence} \\ \text{FC} & \text{False Crawl} \end{cases}$$

As mentioned in the previous chapter, the presence and false crawl data were human observed while the false crawl data was computer generated. For presence, the locations were recorded by a FWC employee on “turtle patrol” and drove the beach, noticing the presence of a nest. The false crawl data was also recorded by a FWC employee who noticed turtle tracks leading up the beach to no nest with a turn around point, leading back to the water. The pseudo absence data was randomly generated using ArcGIS by defining a border around an existing nest, then randomly selecting from these “no known nest” areas.

Beach slope and foreshore slope are continuous predictors calculated based on values for the given location. Beach slope (deg.) is measured from the mean low water line (MLWL) to the mean high water line (MHWL). Foreshore slope (deg.) is measured from the mean high water line (MHWL) to the potential for vegetation. For either slope, a larger value indicates a steeper slope, while a smaller value indicates a flatter slope.

2.2 Statistical Methods

2.2.1 Binary Logistic Regression

Binary logistic regression is used to predict the probability of an outcome when there are only two possible outcomes (9–12). For our purposes, we used binary logistic regression to predict the probability of a nest being present (1) or not (0). Our outcomes were either that a detected nest was present (P), or there was not a nest detected (PA or FC). In order to model binary outcomes using logistic regression, the model will have the following form (9,12):

$$\ln\left(\frac{\hat{\pi}}{1-\hat{\pi}}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + ... + \hat{\beta}_k x_k, \tag{2.1}$$

where

- $\pi = P[Y = 1]$ is the probability of the outcome or event,
- β_0 is the y-intercept,
- β_i is the slope of the i predictor,
- x_i is the value of the i predictor, and
- k is the number of predictors in the model.

Equation 2.1 gives the log-odds of the outcome (observed nest). In order to interpret the regression coefficients ($\hat{\beta}_i$), we must exponentiate. This allows us to interpret the odds directly, rather than the log-odds. The odds ratio, or the exponentiation of the slope, for the i^{th} predictor is:

$$\text{OR}_i = e^{\hat{\beta}_i}. \tag{2.2}$$

Applying Equation 2.2 to the y-intercept ($\exp\{\hat{\beta}_0\}$) gives the baseline odds for a turtle to nest. In other words, this is the odds ratio of a nest ocurring at a given location if all predictors are 0; in this project, that means both beach slope and foreshore slope are flat.

Usually we are interested in comparing β_i against 0 – if a slope is 0, then the predictor has no effect on the outcome as the slope of the line is flat. When dealing with odds ratios, that means we are comparing against 1. If $\exp\{\hat{\beta}_i\} < 1$, the predictor has a negative effect on outcome. If $\exp\{\hat{\beta}_i\} = 1$, the predictor is not related to the outcome. If $\exp\{\hat{\beta}_i\} > 1$, the predictor has a positive effect on the odds of the outcome.

Interpretations for odds ratios are multiplicative. For continuous predictors, we provide the following interpretations for odds ratios: for every 1 unit increase in the predictor, the odds of our outcome are multiplied by $\exp\{\hat{\beta}_i\}$.

To test for the significance of our predictors, we use the Wald z (9,12). The hypotheses are as follows:

- $H_0 : \hat{\beta}_i = 0$
- $H_\text{A} : \hat{\beta}_i \neq 0$

The null hypothesis, H_0, suggests the slope of the probability of the predictor is 0, or flat. The alternative hypothesis, H_A, indicates the slope of the probability of the predictor is not equal to 0. In order for the predictor to be significant, we need to have reason to reject the null hypothesis and conclude the slope is not equal to 0, or not flat.

Once we have established our hypotheses, we then calculate the test statistic using the following formula (9,12):

$$z_0 = \frac{\hat{\beta}_i}{\mathrm{SE}_{\beta_i}}, \tag{2.3}$$

where

- $\hat{\beta}_i$ is the slope of the i predictor, and
- SE_{β_i} is the standard error of β_i.

Using Equation 2.3, we can then find the associated two-sided p-value. If the p-value is less than our significance level, α, we reject H_0 and conclude that the predictor is significant.

Our confidence intervals will be calculated using the following formula (9,12):

$$\hat{\beta}_i \pm z_{\alpha/2} \times \mathrm{SE}_{\beta_i}, \tag{2.4}$$

where

- $\hat{\beta}_i$ is the slope of the i predictor,
- $z_{\alpha/2}$ is the critical value of z, and
- SE_{β_i} is the standard error of β_i.

To find the 95% CI for the OR, we exponentiate Equation 2.4, the confidence interval for β,

$$\exp\{\hat{\beta}_i \pm z_{\alpha/2} \times \mathrm{SE}_{\beta_i}\}. \tag{2.5}$$

2.2.2 Multinomial Logistic Regression

Multinomial logistic regression is used to predict the probability of an event when there are three or more possible unordered outcomes (9–12). If there are c classes in the outcome, this approach simultaneously estimates $c-1$ models. The c^{th} category is used as the reference category while the other $c-1$ categories are compared to it.

The model of multinomial logistic regression is in the form (9,12):

$$\ln\left(\frac{\hat{\pi}_j}{\hat{\pi}_{\text{ref}}}\right) = \hat{\beta}_{0j} + \hat{\beta}_{1j}x_1 + ... + \hat{\beta}_{kj}x_k, \tag{2.6}$$

where $\hat{\pi}_j$ is the probability of the response category, $\hat{\pi}_{\text{ref}}$ represents the probability of the reference category, and k is the number of predictors in the model.

Like in binary logistic regression (Equation 2.1), this initial multinomial logistic model will give us the log-odds of our outcome, an observed nest. We will again exponentiate and interpret in terms of the odds ratio, like in Equation 2.2,

$$\text{OR}_i = e^{\hat{\beta}_{ij}}. \tag{2.7}$$

Odds ratios for multinomial logistic regression models (Equation 2.6) generally interpret the same as in binary logistic regression (Equation 2.1), with a slight change to the comparison in the outcome (10). For continuous predictors, we provide the following interpretations for odds ratios in multinomial logistic regression: for every 1 unit increase in the predictor, the odds of group j in the outcome, as compared to the reference group in the outcome, are multiplied by $\exp\{\hat{\beta}_{ij}\}$.

Statistical significance is performed similarly as in the binary logistic regression case. Both the Wald z from Equation 2.3 and the confidence interval for β_i from Equation 2.4 extend to the multinomial case (9,12). As in binary logistic regression, to find the 95% CI for the OR, we exponentiate the confidence interval for $\hat{\beta}$, as seen in Equation 2.5.

2.2.3 Model Comparison

Both binary logistic and multinomial logistic regression models can be used to predict categorical outcomes. In order to compare the models in this particular application, we will look at classification accuracy and other related measures.

First, we find the predicted probabilities for each observation. In the case of binary logistic regression, we find the predicted probability of a nest being present. We define $\hat{\pi}$ = P[Nest Present]. Mathematically,

$$\widehat{\text{Nested}} = \begin{cases} 1 \text{ (predicted nest)} & \text{if } \hat{\pi} \geq 0.5 \\ 0 \text{ (not a predicted nest)} & \text{if } \hat{\pi} < 0.5 \end{cases}$$

In the case of multinomial logistic regression, we have three possible outcomes, so we find the predicted probabilities for each of the three outcomes. To classify, we look at the predicted probabilities for each observation and assign the observation to the category with the highest predicted probability. We define $\hat{\pi}_{\text{P}}$ = P[Nest Present], $\hat{\pi}_{\text{PA}}$ = P[Pseudo Absence], and $\hat{\pi}_{\text{FC}}$ = P[False Crawl]. Mathematically,

$$\widehat{\text{Nest Type}} = \begin{cases} \text{P} & \text{if } \hat{\pi}_{\text{P}} = \max(\hat{\pi}_{\text{P}}, \hat{\pi}_{\text{PA}}, \hat{\pi}_{\text{FC}}) \\ \text{PA} & \text{if } \hat{\pi}_{\text{PA}} = \max(\hat{\pi}_{\text{P}}, \hat{\pi}_{\text{PA}}, \hat{\pi}_{\text{FC}}) \\ \text{FC} & \text{if } \hat{\pi}_{\text{FC}} = \max(\hat{\pi}_{\text{P}}, \hat{\pi}_{\text{PA}}, \hat{\pi}_{\text{FC}}) \end{cases}$$

Then, to examine the accuracy of the models, we look at confusion matrices and classifications. The confusion matrix shows the observed outcomes and the predicted outcomes. A confusion matrix is defined as follows,

	Observed Positive	Observed Negative	
Predicted Positive	True Positive (TP)	False Positive (FP)	Total Predicted Positives (TP + FP)
Predicted Negative	False Negative (FN)	True Negative (TN)	Total Predicted Negatives (FN + TN)
	Total True Positives (TP + FN)	Total True Negatives (FP + TN)	Total Observations (TP + FP + TN + FN)

The current analysis is interested in comparing the performance of pseudo absence points and false crawl points when representing the lack of an observed nest. Thus, we define a "positive" outcome as an absent nest (PA, FC) and a "negative" outcome as an observed nest (P). Then using the confusion matrix, we can calculate the following measures to assess model performance:

Accuracy

The accuracy of a model is defined as the percentage of the observations were correctly categorized by the model (10,11). In the context of our analysis, accuracy represents the percentage of nests that were correctly classified as present (P) or not present (either PA or FC) by the model. Mathematically,

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{2.8}$$

where

- TP = number of true positives,
- TN = number of true negatives,
- FP = number of false positives, and
- FN = number of false negatives.

Sensitivity

The sensitivity of a model is the probability of correctly identifying positive outcomes (10,11). In the context of our analysis, sensitivity represents the probability of correctly identifying present nests (P) by the model. Mathematically,

$$\text{sensitivity} = \frac{\text{TP}}{\text{TP + FN}} \tag{2.9}$$

where

- TP = number of true positives and
- FN = number of false negatives

Specificity

The specificity of a model is the probability of correctly identifying negative outcomes (10,11). In the context of our analysis, specificity represents the probability of correctly identifying no nest (either PA or FC) by the model. Mathematically,

$$\text{specificity} = \frac{\text{TN}}{\text{FP + TN}} \tag{2.10}$$

where

- TP = number of true positives,
- TN = number of true negatives,
- FP = number of false positives, and
- FN = number of false negatives.

Positive Predictive Value (PPV)

The positive predictive value (PPV) of a model is the probability that a positive prediction is actually positive (10,11). In the context of our analysis, PPV represents the probability that a nest predicted as present (P) is actually present. Mathematically,

$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2.11}$$

where

- TP = number of true positives,
- TN = number of true negatives,
- FP = number of false positives, and
- FN = number of false negatives.

Negative Predictive Value (NPV)

The negative predictive value (NPV) of a model is the probability that a negative prediction is actually negative (10,11). In the context of our analysis, NPV represents the probability that a nest predicted as not present (either PA or FC) is actually not present. Mathematically,

$$\text{NPV} = \frac{\text{TN}}{\text{TN} + \text{FN}} \tag{2.12}$$

where

- TP = number of true positives,
- TN = number of true negatives,
- FP = number of false positives, and
- FN = number of false negatives.

2.2.4 Current Analysis

Binary logistic regression was used to construct two separate models comparing the probability of nesting against the probability of not nesting. We defined not nesting in two different ways, once using pseudo absence points and once using false crawl points. First, we compare presence versus pseudo absence:

$$\ln\left(\frac{\hat{\pi}_{\text{P}}}{\hat{\pi}_{\text{PA}}}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_1 x_2, \tag{2.13}$$

where

- $\hat{\pi}_{\mathrm{P}}$ = probability of presence,
- $\hat{\pi}_{\mathrm{PA}} = 1 - \hat{\pi}_{\mathrm{P}}$ = probability of pseudo absence,
- x_1 = beach slope, and
- x_2 = foreshore slope.

Then, we compare presence versus false crawl:

$$\ln\left(\frac{\hat{\pi}_{\mathrm{P}}}{\hat{\pi}_{\mathrm{FC}}}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_1 x_2, \tag{2.14}$$

where

- $\hat{\pi}_{\mathrm{P}}$ = probability of presence,
- $\hat{\pi}_{\mathrm{FC}} = 1 - \hat{\pi}_{\mathrm{P}}$ = probability of false crawl,
- x_1 = beach slope, and
- x_2 = foreshore slope.

Multinomial logistic regression was used to model all three possible outcome categories simultaenously. Again, we compared presence points against pseudo absence points and, simultaneously, presence points against false crawl points. Mathematically, we simultaneously estimated the following two models:

$$\begin{aligned}\ln\left(\frac{\hat{\pi}_{\mathrm{P}}}{\hat{\pi}_{\mathrm{PA}}}\right) &= \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_1 x_2,\\ \ln\left(\frac{\hat{\pi}_{\mathrm{P}}}{\hat{\pi}_{\mathrm{FC}}}\right) &= \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_1 x_2,\end{aligned} \tag{2.15}$$

where

- $\hat{\pi}_{\mathrm{P}}$ = probability of presence,
- $\hat{\pi}_{\mathrm{PA}}$ = probability of pseudo absence,
- $\hat{\pi}_{\mathrm{FC}}$ = probability of false crawl,
- x_1 = beach slope, and
- x_2 = foreshore slope.

For all models, adjusted odds ratios, 95% confidence intervals for the adjusted odds ratios, and significance of predictors are reported. Statistical significance was defined *a priori* as $p < 0.05$.

Predicted values were calulated based on estimated models, then compared against true values, constructing the confusion matrix. Modeling approaches are compared by considering the resulting classification accuracy, sensitivity of nest detection, specificity of nest detection, positive predictive value (PPV), and negative predictive value (NPV) of each model.

Data management, analysis, and graphing was performed using R version 4.5.1 (13). Data management was completed using functions from `tidyverse` package (14). The `glm` function was used to perform binary logistic regression, while the `multinom` function from the `nnet` package was used to perform multinomial logistic regression (15). Finally, the `caret` package was used to create confusion matrices (16).

2.3 Results

2.3.1 Estimated Models

Binary Logistic Regression

Presence vs. False Crawl

Modeling the probability of the presence of a nest (1) versus a false crawl (0), we looked at nest status as a function of beach slope, foreshore slope and the interaction between beach slope and foreshore slope. The model produced was:

$$\ln\left(\frac{\hat{\pi}_{\mathrm{P}}}{\hat{\pi}_{\mathrm{FC}}}\right) = 0.61 + 0.016x_1 + 0.046x_2 - 0.0079x_1x_2, \tag{2.16}$$

where

- $\hat{\pi}_{\mathrm{P}}$ = probability of presence,
- $\hat{\pi}_{\mathrm{FC}}$ = probability of false crawl,
- x_1 = beach slope, and
- x_2 = foreshore slope.

Because the interaction between beach slope and foreshore slope was not significant, removed the interaction from the model for parcimony and interpretability. The model without the interaction term was:

$$\ln\left(\frac{\hat{\pi}_{\mathrm{P}}}{\hat{\pi}_{\mathrm{FC}}}\right) = 0.88 + 0.049x_1 + 0.011x_2, \tag{2.17}$$

where

- $\hat{\pi}_{\mathrm{P}}$ = probability of presence,
- $\hat{\pi}_{\mathrm{FC}}$ = probability of false crawl,
- x_1 = beach slope, and
- x_2 = foreshore slope.

These results reveal that for a 1 degree increase in the beach slope, the odds of a nest being present are multiplied by $e^{-0.049} = 0.95$, or are decreased by 5%. When foreshore slope increases by 1 meter, the odds of a nest being present are multiplied by $e^{0.011} = 1.01$, or are increased by 1%.

Presence vs. Pseudo Absence

For the probability of the presence of a nest (1) versus a pseudo absence (0) we also considered the interaction model where nest status was modeled as a function of beach slope, foreshore slope and the interaction between beach slope and foreshore slope. The model produced was:

$$\ln\left(\frac{\hat{\pi}_{\mathrm{P}}}{\hat{\pi}_{\mathrm{PA}}}\right) = 0.34 + 0.013x_1 + 0.009x_2 - 0.0001x_1x_2, \tag{2.18}$$

where

- $\hat{\pi}_{\mathrm{P}}$ = probability of presence,
- $\hat{\pi}_{\mathrm{FC}}$ = probability of false crawl,
- x_1 = beach slope, and
- x_2 = foreshore slope.

The interaction between beach slope and foreshore slope was again not significant ($p > 0.05$), it was removed for parcimony. The model without the interaction term was:

$$\ln\left(\frac{\hat{\pi}_{\mathrm{P}}}{\hat{\pi}_{\mathrm{PA}}}\right) = 0.31 + 0.022x_1 + 0.013x_2 \tag{2.19}$$

where

- $\hat{\pi}_{\mathrm{P}}$ = probability of presence,
- $\hat{\pi}_{\mathrm{FC}}$ = probability of false crawl,
- x_1 = beach slope, and
- x_2 = foreshore slope.

These results reveal that for a 1 degree increase in the beach slope, the odds of a nest being present are multiplied by $e^{0.022} = 1.02$, or are increased by 2%. For a 1 degree increase in foreshore slope, the odds of a nest being present are multiplied by $e^{0.013} = 1.01$, indicating a 1% increase in the odds.

Multinomial Logistic Regression

For multinomial logistical regression models we looked at the same model structures, but with the model accounting for all three possible outcomes simultaneously. First looking at the models including the interation between beach slope and foreshore slope:

$$\begin{aligned}
\ln\left(\frac{\hat{\pi}_{\mathrm{P}}}{\hat{\pi}_{\mathrm{PA}}}\right) &= -0.569 + 0.029x_1 + 0.026x_2 + 0.007x_1x_2 \\
\ln\left(\frac{\hat{\pi}_{\mathrm{P}}}{\hat{\pi}_{\mathrm{FC}}}\right) &= \quad 0.637 + 0.008x_1 + 0.041x_2 - 0.007x_1x_2
\end{aligned} \tag{2.20}$$

As in the binary logistic regression models, the interaction between beach slope and foreshore slope was not significant ($p > 0.05$). The interaction term was removed for parcimony, resulting in the following models:

$$\begin{aligned}
\ln\left(\frac{\hat{\pi}_{\mathrm{P}}}{\hat{\pi}_{\mathrm{PA}}}\right) &= -0.759 + 0.088x_1 + 0.051x_2 \\
\ln\left(\frac{\hat{\pi}_{\mathrm{P}}}{\hat{\pi}_{\mathrm{FC}}}\right) &= \quad 0.890 + 0.048x_1 + 0.010x_2
\end{aligned} \tag{2.21}$$

In the first model, with pseudo-absence points as the reference category, we see that for a 1 degree increase in the beach slope, the odds of a nest being present, as compared to a pseudo absence, are multiplied by $e^{0.088} = 1.09$, indicating a 9% increase in odds. For a 1 degree increase in foreshore slope, the odds of a nest being present, as compared to a pseudo absence, are multiplied by $e^{0.051} = 1.05$, which shows a 5% increase in odds.

In the second model, with false crawl points as the reference category, we see that for a 1 degree increase in the beach slope, the odds of a nest being present, as compared to a false crawl, are multiplied by $e^{0.048} = 0.95$, indicating a 5% decrease in odds. For a 1 degree increase in foreshore slope, the odds of a nest being present, as compared to a pseudo absence, are multiplied by $e^{0.0095} = 1.01$, which shows a 1% increase in odds.

2.3.2 Classification

Binary Logistic Regression

Considering only the models predicting nesting status as a function of both beach slope and foreshore slope, confusion matrices were examined to determine the accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) for each model.

Table 2.2 shows the confusion matrix for the binary logistic regression model predicting presence versus pseudo absence, while Table 2.3 shows the confusion matrix for the binary logistic

Table 2.2: Binary Logistic Regression: Pseudo Absence Confusion Matrix

	Observed	
	PA	P
Predicted		
PA	51	27
P	41	38

Table 2.3: Binary Logistic Regression: False Crawl Confusion Matrix

	Observed	
	FC	P
Predicted		
FC	0	38
P	0	79

Table 2.4: Multinomial Logistic Regression: Confusion Matrix

Observed
P
PA
FC
Predicted
P
38
27
20
PA
41
51
18
FC
0
0
0

regression model predicting presence versus false crawl. The model comparing against pseudo absence correctly classified 38/79 presence points and 51/78 pseudo absence points while the model comparing against false crawl correctly classified 79/79 presence points and 0/38 false crawl points.

Multinomial Logistic Regression

Examining the confusion matrix for the multinomial logistic regression model predicting among presence, pseudo absence, and false crawl, means that we now have a 3×3 confusion matrix to consider.

Table 2.4 shows the confusion matrix for the multinomial logistic regression model predicting among presence, pseudo absence, and false crawl. The model correctly classified 38/85 presence points, 51/110 pseudo absence points, and 0/0 false crawl points.

In order to compare the models side by side, we will now consider the "positive" outcome as either PA vs. P+FC or FC vs. P+PA for the multinomial logistic regression model. This allows us to reduce the 3×3 confusion matrix into a 2×2 confusion matrix.

Table 2.5 and Table 2.5 show the confusion matrices for the multinomial logistic regression model predicting presence versus pseudo absence, The model comparing against pseudo absence correctly classified 38/79 presence points and 51/78 pseudo absence points while the model comparing against false crawl correctly classified 79/79 presence points and 0/38 false crawl points.

Table 2.5: Multinomial Logistic Regression: Pseudo Absence Confusion Matrix

	Observed	
	PA	P or FC
Predicted		
PA	51	59
P or FC	27	58

Table 2.6: Multinomial Logistic Regression: False Crawl Confusion Matrix

	Observed	
	FC	P or PA
Predicted		
FC	0	0
P or PA	38	157

Table 2.7: Model Performance: Presence vs. Pseudo Absence

Modeling Approach	Predictor	OR (95% CI)	*p*-value
Binary	Foreshore Slope	1.01 (0.98 - 1.05)	0.460
	Beach Slope	1.02 (0.99 - 1.05)	0.154
Multinomial	Foreshore Slope	1.05 (0.92 - 1.21)	0.467
	Beach Slope	1.09 (0.97 - 1.23)	0.162

2.3.3 Model Comparison

Model Results and Interpretations

Table 2.7 shows model results for the binary and multinomial logistic regressions using pseudo absence as the reference group while Table 2.8 shows the same but using false crawl as the reference group.

Comparing the two modeling approaches, we see that slopes, confidence intervals, and p-values are similar when comparing observed nests to pseudo absence points. For foreshore slope, the adjusted odds ratio under binary logistic is 1.01 (95% CI: 0.98, 1.05; $p = 0.460$) and is 1.05 (95% CI: 0.92, 1.21; $p = 0.467$) under multinomial logistic. Similarly, the adjusted odds ratio for beach slope under binary logistic is 1.02 (95% CI: 0.99, 1.05; $p = 0.154$) and is 1.09 (95% CI: 0.97, 1.23; $p = 0.162$) under multinomial logistic.

The corresponding interpretations are similar as well. For a 1 degree increase in foreshore slope, the odds of nesting increase by 1% under the binary logistic model and by 5% under the multinomial logistic model. For a 1 degree increase in beach slope, the odds of nesting increase by 2% under the binary logistic model and by 9% under the multinomial logistic model.

Again comparing the two modeling approaches, we see that slopes, confidence intervals, and p-values are similar when comparing observed nests to false crawl points. For foreshore slope, the adjusted odds ratio under binary logistic is 1.01 (95% CI: 0.85, 1.20; $p = 0.899$) and is also

Table 2.8: Model Performance: Presence vs. False Crawl

Modeling Approach	Predictor	OR (95% CI)	p-value
Binary	Foreshore Slope	1.01 (0.85 - 1.20)	0.899
	Beach Slope	0.95 (0.83 - 1.10)	0.487
Multinomial	Foreshore Slope	1.01 (0.85 - 1.20)	0.914
	Beach Slope	0.95 (0.83 - 1.09)	0.490

1.01 (95% CI: 0.85, 1.20; $p = 0.914$) under multinomial logistic. Similarly, the adjusted odds ratio for beach slope under binary logistic is 0.95 (95% CI: 0.83, 1.10; $p = 0.487$) and is 0.95 (95% CI: 0.83, 1.09; $p = 0.490$) under multinomial logistic.

The corresponding interpretations are the same between the two models. For a 1 degree increase in foreshore slope, the odds of nesting increase by 1% under both the binary and multinomial logistic models. For a 1 degree increase in beach slope, the odds of nesting decrease by 5% under the binaryand multinomial logistic models.

Model Performance

Confusion matrices were be used to assess the performance of each model. The following measures were calculated for each model: accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV). Table 2.9 shows the performance measures for each of the four models considered.

We see that the binary model accurately categorized 57% of the pseudo absence points and 68% of the false crawl points. The sensitivity shows that the binary model is correctly identifying 55% of pseudo absence points, however, the model never categorized an observation as a false crawl. Therefore, we cannot compute sensitivity when comparing present nests to false crawl. The specificity shows the model correctly identifies observations as pseudo absence 58% of the time and false crawl 68% of the time. The positive predictive value shows that out of what was

Table 2.9: Model Performance

	Binary		**Multinomial**	
	PA	**FC**	**PA**	**FC**
Accuracy	0.57	0.68	0.56	0.81
Sensitivity	0.55	–	0.65	0.00
Specificity	0.58	0.68	0.50	1.00
PPN	0.65	0.00	0.46	–
NPV	0.48	1.00	0.68	0.81

categorized as pseudo absence points, 65% were actually pseudo absence points and 0% were actually false crawl points. For the negative predictive value, the model correctly identifying observed nests 48% of the time when compared to pseudo absence points and 100% of the time when compared to false crawl points.

The results of the multinomial logistic regression are somewhat similar to those of the binary logistic regression. Like before, when false crawl is the reference group of the outcome, the model never predicts an observation is a false crawl.

The multinomial model accurately categorized 56% of the pseudo absence points and 81% of the false crawl points. The sensitivity shows that the multinomial model is correctly identifying 65% of pseudo absence points, however, the model never categorized an observation as a false crawl, making the sensitivity 0%. The specificity shows the model correctly identifies observations as pseudo absence 50% of the time and false crawl 100% of the time. The positive predictive value shows that out of what was categorized as pseudo absence points, 46% were actually pseudo absence points; because no nests were categorized as false crawls, the positive predictive value cannot be computed. For the negative predictive value, the model correctly identified nests 68% of the time when comparing against pseudo absence points and 81% of the time when compared to false crawl points.

2.3.4 Model Visualization

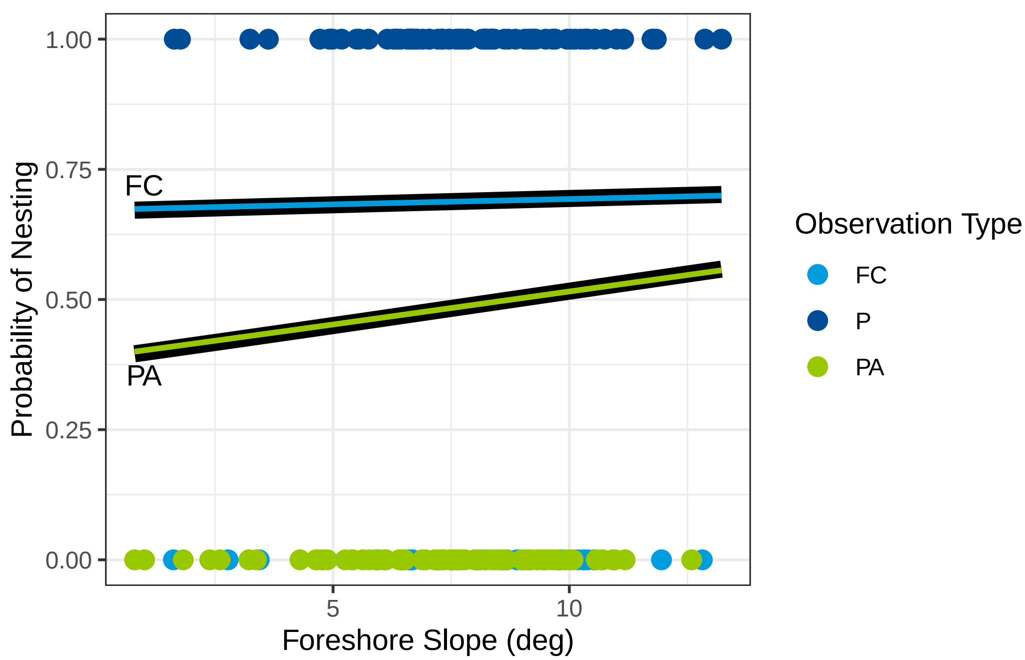

Figure 2.1: Model Visualization: Probability of Nesting vs. Foreshore Slope

In the above data visualization, the binary logistic regression models are represented as the thick black lines. The multinomial logistic regression models are the thin blue and green lines overlaid. As can be seen, the resulting models are not different in a meaningful way. While the models aren't exactly the same, the predicted probabilities are very similar between the two modeling approaches. This further supports the findings from the model results and performance measures that both modeling approaches yield similar results.

3 Conclusions

There was no statistical relationship between sea turtle nesting on Pensacola Beach and either foreshore slope or beach slope. It is possible that the nesting habits of loggerhead sea turtles are influenced by factors not included in our study and unavailable in our dataset.

When comparing the two modeling approaches, binary logistic regression and multinomial logistic regression, we also found no meaningful difference. Both methods resulted in similar conclusions regarding the relationship between nesting status and the beach characteristics. While the regression coefficients are different in the resulting models, they do not differ in a meaningful way. The story of the data remains the same, regardless of the modeling approach.

When directly comparing the use of false crawl points versus pseudo absence points as the “no nest” observation, we found no meaningful difference between the two methods. Both methods struggled to accurately classify the nesting status of the locations, especially false crawls. None of the modeling approaches predicted a single false crawl point. Thus, we saw low accuracy, sensitivity, and specificity. This is likely due to the lack of a relationship between nesting status and the beach characteristics as well as a small sample size.

3.1 Limitations

There are several limitations in this study and analysis.

First, we are limited by the data collection methods. This analysis relies on a combination of human collected data and randomly generated data from ArcGIS. For the observed nests and observed false crawls, FWC employees recorded information during turtle patrol; human error will inherently play a role. The pseudo absence points are randomly generated by ArcGIS, profile lines are then created by our environmental science collaborators, then beach slope and foreshore slope are calculated from those profile lines.

Next, there is not a large population of loggerhead sea turtles nesting on Pensacola Beach. This leads to a smaller sample size, limiting the statistical power of our analysis, and results may not be generalizable outside of Escambia County. Similarly, this analysis is only for one nesting season and may not be representative of other years.

The true relationship between loggerhead sea turtle nesting status and either the beach slope or foreshore slope of the location is unknown. In order to truly compare the methods, a full simulation study is needed so that bias and mean squared error can be examined as well.

3.2 Suggestions for Further Study

Limitations can be addressed in future studies. While data collection methods cannot be changed, the Computational Geomorphology and Modeling Lab plans to expand upon this project.

First, more data will be collected over multiple nesting seasons to increase the sample size and statistical power of the analysis, adding a temporal component to the analysis. Similarly, we can expand the spatial study area to include other beaches in the Florida Panhandle or throughout the State of Florida. This would increase the generalizability of the results.

A full Monte Carlo simulation study can also be conducted to truly compare the methods. By specifying the true relationship between nesting status and the beach characteristics, we can evaluate bias and mean squared error of the resulting models. This will provide more insight into which method is “better” in terms of estimating the true relationship.

Finally, other environmental predictors can be included in the analysis, if observed and available. There are many factors that may influence sea turtle nesting behavior, such as sand temperature, vegetation, human activity, and artificial lighting. Including additional predictors may help to better understand the nesting habits of loggerhead sea turtles on Pensacola Beach.

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